

Route-F P54: Common Scalar Spectrum, Interaction-Defined Channels, and Source-Preserving Nested Elimination

P54PQ-v2 four-dimensional mainline

10 September 2026

Abstract

We keep the existing action and stationary benchmark fixed and construct tree scalar-exchange form factors from actual Higgs, Yukawa and vector vertices. The complete physical scalar space has 295 real dimensions; 33 declared real current slots generate a 21-dimensional invariant spectral subspace, with 12 visible eigenvalue clusters. This is a channel-dependent exact reduction, not an additional spatial dimension or a new particle fit. A precise Gaussian elimination law transports the kernel, vertices and induced source-source term together. Omitting the last term loses the entire Higgs-pair scalar-exchange response in the tested final chart. An initial nonzero NN -PQ coupling conjecture is rejected: the physical PQ tangent is the gauge-singlet phase and has no linear residue in the declared channels. All 153/153 checks pass without recomputing cached Hessians or changing parameters.

1 Scope and unchanged theory

The frozen action remains P54PQ-v2, with

$$3\,16_F + 54_{H,\mathbb{R}} + 126_{H,\mathbb{C}} + 10_{H,\mathbb{C}} + 1_{H,\mathbb{C}}, \quad (1)$$

and only two symmetric UV family matrices $h_{\text{raw}}, f_{\text{raw}}$. We reuse its 328-real-coordinate polynomial and the stationary benchmark $(\omega, \sigma, v_s) = (1, 0.1265, 0.25)$ in program units. The gauge coupling is read from the historical P2 iteration, with the reference convention $\mu_U = g_U \omega$. Neither its dimensionful unification scales nor the benchmark’s phenomenological viability are recertified here.

The primary Higgs direction is the *exact tree* light eigenvector. The bosonic-improved vector is not substituted into a claimed tree or loop-complete amplitude. All numerical Yukawa residues below are coefficients with $h_{\text{raw}}, f_{\text{raw}}$ factored out. Fermion masses, flavor mixing and rates are not fitted. The result is a set of amputated tree *scalar-exchange* kernels, not a complete gauge-invariant scattering calculation including vector exchange, contact diagrams, loops and external-leg matching.

The conceptual motivation is the proposed “visible energy spectrum” in Ref. [1]. Here visibility is defined by actual interaction vertices, not an arbitrarily chosen geometrical projection. Different description layers do not imply different numbers of spatial dimensions. No mechanical rotation, helical orbit or particle-specific geometry is introduced.

2 Physical scalar spectrum and interaction vertices

2.1 Remove gauge directions, retain physical zero modes

Let $H = V^{(2)}(x_0)$ and let G be an orthonormal basis of the rank-33 broken gauge orbit $\{R_A x_0\}$. At a stationary vacuum, gauge invariance gives $HG = 0$. Choose an orthonormal physical slice B such that

$$BB^T = P_{\text{phys}} = I - GG^T, \quad B^T B = I_{295}, \quad H_p = B^T H B. \quad (2)$$

The full physical tree spectrum has 290 positive eigenvalues and five zeros: four real Higgs components and one physical PQ mode. A zero eigenvalue is not by itself a reason to discard a state. Conversely, eaten Goldstones are not additional physical scalar particles.

For orthogonal eigenspace projectors Π_λ of H_p ,

$$H_p = \sum_\lambda \lambda \Pi_\lambda, \quad \sum_\lambda \Pi_\lambda = I_{295}. \quad (3)$$

For canonical free propagation about this vacuum, the corresponding energy branches are $E_\lambda(\mathbf{p}) = \sqrt{|\mathbf{p}|^2 + \lambda}$ in natural units. Thus mass is the rest-energy gap of a branch, not an additional coordinate or an observer-selected energy value. There are 24 distinct numerical tree eigenvalue clusters, including zero. This count is not the earlier 35-row SM-irrep census: distinct irreps can share a mass, and gauge directions have now been removed.

2.2 Higgs-pair source from the same potential

Let q_α , $\alpha = 1, \dots, 4$, be the canonical real components of the unique tree light doublet, and $\rho = H_{\text{light}}^\dagger H_{\text{light}} = \frac{1}{2} \sum_\alpha h_\alpha^2$. For SM-singlet scalar fluctuations s_A , the potential contains

$$V \supset \frac{1}{2} s^T K s + s^T J_\rho \rho, \quad (J_\rho)_A = \frac{1}{4} \sum_{\alpha=1}^4 V^{(3)}[e_A, q_\alpha, q_\alpha]. \quad (4)$$

Equivalently $J_\rho = P_{\text{SM singlet}} V^{(3)}[\cdot, q, q]$ for one canonical neutral component q . We verify this equality by the full four-component trace, not by selecting a neutral triplet as a detector. Differentiating the gauge identity twice gives

$$(R x_0)^T V^{(3)}[\cdot, q, q] + 2(R q)^T H q = 0. \quad (5)$$

Thus the exact tree condition $H q = 0$ eliminates gauge-orbit contamination of this source. The cubic jets are recovered from content-addressed Hessian differences; a central difference of the Hessian of a quartic polynomial is exact algebraically. No new lattice or parameter scan is performed.

2.3 Yukawa sources and their selection rules

In the fixed common complex scalar/spinor convention,

$$M_F(x) = h_{\text{raw}} \otimes \sum_A Z_h^A x_A + f_{\text{raw}} \otimes \sum_A Z_f^A x_A. \quad (6)$$

For the declared spinor components l, r , define

$$g_{A,h}^{lr} = l^T Z_h^A r, \quad g_{A,f}^{lr} = l^T Z_f^A r. \quad (7)$$

The real and imaginary parts label Hermitian-current components; the actual family matrices multiply the corresponding bilinears and are not new independent fit parameters for different channels. The seven component pairs are $u_L u^c, d_L d^c, e_L e^c, \nu_L N, N N, \nu_L \nu_L, u^c d^c$. Quark pairs use the first matched colour of the common-phase dictionary; these are not colour-averaged bilinears. Zero components are retained as explicit selection-rule tests.

For an unbroken generator the full tensor identity is

$$R_F^T Z^A + Z^A R_F + \sum_B Z^B (R_s)_{BA} = 0. \quad (8)$$

This is checked for both tensors and all 12 SM generators. For hypercharge eigenstates l, r with charges y_l, y_r , it implies, in the real scalar coordinates used here,

$$i R_Y g^{lr} = (y_l + y_r) g^{lr}. \quad (9)$$

This defines allowed support independently of the scalar eigenvalues. In particular $g_h^{NN} = g_h^{\nu_L \nu_L} = 0$, whereas the corresponding 126 vertices are nonzero. A zero for a selected coloured component must not be generalized to all colour contractions.

2.4 Vector-pair sources from the kinetic term

With $D = \partial + gA_a R_a$ and canonical real fields,

$$(M_V^2)_{ab} = g^2 (R_a x_0)^T (R_b x_0). \quad (10)$$

For a complete degenerate vector eigenspace with projector P_β , dimension d_β and corresponding generators $\tilde{R}_b$, the averaged scalar–vector–vector vertex is

$$(J_\beta)_A = \frac{1}{d_\beta} \text{tr} \left(P_\beta \frac{\partial M_V^2}{\partial x_A} \right) = \frac{2g^2}{d_\beta} \sum_{b \in \beta} e_A^T \tilde{R}_b^T \tilde{R}_b x_0. \quad (11)$$

It multiplies the metric contraction of the two external vector polarizations; it is not yet a decay rate. The four positive mass eigenspaces have real multiplicities 8, 1, 12, 12 and mass squares 0.00506506, 0.02532532, 0.31652201, 0.32158708 in ω^2 . Degeneracy averaging is fixed before the probe is evaluated. Every such source is an SM singlet. For unbroken massless vectors $R_b x_0 = 0$, so this tree single-scalar mass vertex vanishes. An independent finite difference of the kinetic-term mass trace checks its factor of two.

3 Common poles, channel residues and a minimal realization

Collect all physical source columns into $J = B^T J_{\text{raw}}$. In units $\omega = 1$ there are 33 real current slots: one Higgs-pair source, 28 Yukawa slots and four vector-pair sources. Different current dimensions are explicitly recorded; the Yukawa spurions are factored out. At Euclidean $t = p_E^2$ the scalar-exchange response is

$$F_E(t) = J^T (H_p + tI)^{-1} J = \sum_{\lambda} \frac{R_\lambda}{t + \lambda}, \quad R_\lambda = J^T \Pi_\lambda J \succeq 0. \quad (12)$$

Minkowski continuation is $t = -p^2 - i0$. The pole positions are properties of the common operator; the residues record which channels can access each pole. Vanishing residue does not change the spectrum or turn an unseen state into negative mass. These are finite tree spectral-theorem statements, not an all-orders spectral representation for gauge-fixed unphysical fields.

For any real channel vector c , $c^T R_\lambda c = \|\Pi_\lambda J c\|^2 \geq 0$. Degenerate eigenvectors may be relabelled without changing R_λ . The two independent moment checks are

$$\sum_{\lambda} R_\lambda = J^T J, \quad \sum_{\lambda} \lambda R_\lambda = J^T H_p J. \quad (13)$$

The machine-readable artifact retains full cross-channel residues, not only squared diagonal couplings. Relative phases and interference cannot in general be reconstructed from diagonal weights alone.

Proposition 1 (Interaction-generated minimal spectral space). *For finite self-adjoint H_p , let*

$$\mathcal{H}_J = \text{span}\{\text{ran}(\Pi_\lambda J) : \lambda \in \text{spec } H_p\}. \quad (14)$$

Then $\mathcal{H}_J$ is the smallest H_p -invariant subspace containing $\text{ran } J$. Its orthogonal complement is invariant and does not contribute to F_E for this current set. Restricting H_p, J to $\mathcal{H}_J$ reproduces the response exactly at every point of the resolvent set.

Proof. Each generator $\Pi_\lambda J c$ is an eigenvector or zero, so the span is invariant, and $\sum_{\lambda} \Pi_\lambda J = J$ puts $\text{ran } J$ inside it. Conversely, on a finite spectrum each Π_λ is a polynomial in H_p by Lagrange interpolation. Any invariant subspace containing J therefore contains every $\Pi_\lambda J$. Self-adjointness makes its orthogonal complement invariant. In this decomposition J has zero component on the complement, proving the resolvent identity. $\square$

For the declared component currents, rank $J = 19$ but $\dim \mathcal{H}_J = 21$, with 12 visible eigenvalue clusters. Thus a mere projection onto the instantaneous source range loses dynamically accessible directions. The 21-dimensional exact realization is verified against the full 295-dimensional response at five momenta. This is not a universal claim that P54 has only 21 scalar degrees of freedom: different probes or special fitted family cancellations change the observable subspace. In particular the component-current set is not closed under every possible external colour/isospin choice.

4 Actual singlet response and a rejected PQ hypothesis

m^2/ω^2	$R_{\rho\rho}/\omega^2$	$R_{NN}^{\text{unit spurion}}$
0.01352047	2.3534868×10^{-7}	15.947239
0.09971569	0.00020922086	0.049259875
2.82916586	0.07540586	0.0035014756

The last column means the sum of the two real unit-spurion residue components, not a numerical flavor-matrix norm. The same three radial poles appear in both channels but with very different weights. No mass-list fitting was used to produce this result.

4.1 The physical PQ zero mode is not an NN pole here

An initial hypothesis that the NN channel necessarily sees the PQ zero mode failed in the first run. The reason follows from support, not from a numerical tuning. At this vacuum the phase variation of the 126 vacuum lies in the broken gauge orbit. The physical PQ tangent is

$$a = \frac{P_{\text{phys}} \delta_{\text{PQ}} x_0}{\|P_{\text{phys}} \delta_{\text{PQ}} x_0\|}, \quad (I - P_S)a = 0. \quad (15)$$

There is no renormalizable $S16_F 16_F$ term in the frozen action, hence

$$Z_h^A = Z_f^A = 0 \quad (A \in S), \quad g_{NN}^T a = 0. \quad (16)$$

The actual J_ρ and vector-pair vertices are also orthogonal to a . All declared linear scalar residues vanish on this physical mode. The verifier separately checks its support, Yukawa tensor zero, $Ha = 0$ and all current contractions. This does not imply that the axion is unphysical, absent, globally decoupled or massless after QCD effects. Other vertices, derivative descriptions and anomalies are outside the declared tree scalar-exchange current set.

4.2 Recover the existing nonzero sterile-Higgs coefficient as a spectral moment

Let B_s embed the three massive radial modes and define $K_s = B_s^T H B_s$, $j_\rho = B_s^T J_\rho$ and $G_A = (g_{NN}^T B_s)_A f_{\text{raw}}$. Then

$$\mathcal{L} \supset -\frac{1}{2} s^T K_s s - s^T j_\rho \rho - \left[\frac{1}{2} N^T (M_R + G_A s_A) N + \text{h.c.} \right]. \quad (17)$$

Solving the scalar equation at leading order gives $s = -K_s^{-1} j_\rho \rho + \dots$. In the convention $\mathcal{L} \supset C_{HN}^{ij} N_i N_j \rho + \text{h.c.}$, with no 1/2 in this operator, define physical-slice columns $\bar{g} = B^T g_{NN}$ and $\bar{J}_\rho = B^T J_\rho$. Then

$$C_{HN} = \frac{1}{2} G^T K_s^{-1} j_\rho = \frac{f_{\text{raw}}}{2} \sum_{\lambda > 0} \frac{\bar{g}^T \Pi_\lambda \bar{J}_\rho}{\lambda}, \quad (18)$$

$$\omega C_{HN} = -0.0848692011507 i f_{\text{raw}}. \quad (19)$$

This reproduces the prior same-action coefficient and its common component phase. The use of g^T , not $g^\dagger$, is required for this holomorphic Majorana operator. Positivity applies instead to the Hermitian-current residue matrix in Eq. (12). No singular zero-mode inverse is used, and no absolute physical Yukawa rate is claimed.

5 Nested elimination preserves a triple, not just a mass matrix

5.1 Exact Gaussian derivation with external currents

Let $x = (x_r, x_h)$ and let η collect real external current sources. Define the quadratic Euclidean functional

$$S_E[x, \eta] = \frac{1}{2}x^T K x - x^T J \eta - \frac{1}{2}\eta^T C \eta. \quad (20)$$

At its stationary point, the current response is $F = C + J^T K^{-1} J$. Solving the heavy equation gives

$$x_h = K_{hh}^{-1}(J_h \eta - K_{hr} x_r). \quad (21)$$

Completing the square then gives the exact transformation

$$K' = K_{rr} - K_{rh} K_{hh}^{-1} K_{hr}, \quad (22)$$

$$J' = J_r - K_{rh} K_{hh}^{-1} J_h, \quad (23)$$

$$C' = C + J_h^T K_{hh}^{-1} J_h. \quad (24)$$

Subsequent integration of x_r yields

$$\boxed{C + J^T K^{-1} J = C' + J'^T K'^{-1} J'}. \quad (25)$$

This is the source-dependent form of the Feshbach–Schur construction [2, 3]. At complex p_E^2 these are analytic transposes, not Hermitian conjugates of the continued kernel. Hermitian positivity was established on the real spectral residues before continuation.

Proposition 2 (Nested consistency). *For nested retained coordinate sets and invertible eliminated blocks, successive exact transformations of (K, J, C) give the same final triple as simultaneous elimination. Changing the order of the eliminated blocks gives the same response on their common domain of definition.*

Proof. The heavy linear equations have a unique solution for each retained field and source. Successive substitutions and a simultaneous solution therefore give the same heavy stationary value. Substitution into the same quadratic functional fixes separately the coefficients of x_r^2 , $x_r \eta$ and η^2 , which are K', J', C' . This proves the triple identity. Alternatively it follows by block Gaussian elimination on the augmented matrix

$$\begin{pmatrix} K & -J \\ -J^T & -C \end{pmatrix}. \quad (26)$$

At exceptional block singularities one must change the split or use analytic continuation of the full response, not invert a zero mode by an undeclared regulator. $\square$

The induced $C'(p^2)$ is a source-source term with no retained-field propagator, but it is generally *nonlocal* and can contain poles. It becomes a local Wilson expansion only inside a valid momentum domain. Calling it a “contact term” does not mean it is constant or dispensable. In particular an eigenvalue of an eliminated block can produce singularities in individual K', J', C' which cancel in the complete response; it must not automatically be advertised as a new physical pole.

5.2 Actual P54 factorization and negative control

We reuse the upper 55-scalar physical embedding B_U . The final chart B_L contains the actual eight-real-dimensional 10_H bidoublet parent and the physical PQ tangent. The remaining physical coordinates define B_M , giving the exact split

$$295 = 9_L + 231_M + 55_U. \quad (27)$$

This is a coordinate factorization of the same quadratic theory, not an assertion that the nine-coordinate chart is a local SM EFT. The test keeps the full momentum dependence, all sources and the induced source-source functional.

At $p_E^2/\omega^2 = 0.001, 0.03, 0.2, -0.02 + 0.003i, -0.265 + 0.001i$, direct elimination, U then M , and M then U agree. The full response also agrees with its pole-residue sum and with passive noncanonical coordinate rescaling, provided the kinetic metric and vertices are transformed together. These operations are tests of representation covariance, not physical rotations of matter.

The final chart has no massive SM-singlet scalar. Selection rules give $J'_\rho = 0$, although $F_{\rho\rho} \neq 0$. All its scalar-exchange response therefore resides in $C'_{\rho\rho}$. Omitting C' loses 100% of this response at every tested momentum. For the entire normalized current matrix, the analogous relative Frobenius error ranges from about 0.366 to 1. That matrix norm depends on the declared current units; it is not a universal percentage error in a physical cross section. The single-channel $\rho\rho$ negative control needs no fitted Yukawa matrix and does not depend on relative current rescaling.

5.3 What this does not close

An interacting loop-level elimination must transport higher vertices, the functional measure and regulated determinant as well. The Gaussian determinant at this fixed background is source independent; dropping it is harmless for the present tree quadratic response but not for a claimed full effective action or finite matching. The exact identities do not complete the outstanding common gauge/Goldstone/ghost/scalar hard-minus-EFT calculation. A finite derivative truncation preserves them only to its declared order and within its validity domain.

6 Extra spatial dimensions: a separate, constrained comparison

No extra direction is used above. If one independently postulates a flat interval $y \in [0, L]$ and a free real five-dimensional scalar,

$$S = \frac{1}{2} \int d^4x \int_0^L dy \left[(\partial_\mu \Phi)^2 - (\partial_y \Phi)^2 - M_5^2 \Phi^2 \right], \quad (28)$$

Neumann conditions and the self-adjoint eigenproblem give

$$-f_n'' = \lambda_n f_n, \quad f_n'(0) = f_n'(L) = 0, \quad f_0 = L^{-1/2}, \quad f_{n>0} = \sqrt{2/L} \cos(n\pi y/L). \quad (29)$$

Expanding $\Phi = \sum_n \phi_n f_n$ and using orthonormality yields

$$m_n^2 = M_5^2 + (n\pi/L)^2, \quad \frac{m_2^2 - m_0^2}{m_1^2 - m_0^2} = 4. \quad (30)$$

Thus two common parameters predict an entire tower. For a declared probe profile $w(y)$, residues are also fixed by $|\int_0^L w(y) f_n(y) dy|^2$; w may not be chosen separately for each particle. Changing boundary conditions, potentials or particle assignments adds assumptions and must be counted. This conditional construction is not adopted or fitted, and it does not derive chiral families, charges or the observed SM spectrum. The plane-wave/normal-mode energy is not a mechanical helical trajectory.

7 Reproducibility and next gate

The verifier is `code/verify_p54_spectral_channels.py`; its JSON contains all source columns, complete physical mass eigenvalues, residue matrices, selection rules, source hashes and content-addressed cache reads. It never overwrites or recomputes the supplied Hessian cache. Use the pinned `code/requirements_p54_matching.txt` environment:

```
python3 route_f/code/verify_p54_spectral_channels.py \  
--cache-dir /path/to/tmp/p54_full_doublet_cw
```

The matching card, common-phase dictionary and prior C_{HN} output are required inputs. 153/153 tests pass. This count is a computational regression ledger; the propositions state separately the assumptions of the mathematical identities.

The next mainline task is to incorporate the complete source-dependent functional into the same-scheme finite matching workflow. The unknown family matrices must remain correlated inputs until a constrained fit exists. The present scalar tree gate is closed; full finite matching, fermion-inclusive light-state iteration, physical pole widths and a global flavor/seesaw fit remain open. The extra-dimensional comparison and the independent soliton line do not block this sequence.

References

- [1] B. Chen, *Energy spectrum for elementary particles* (2022), six-page manuscript, <https://indico.cern.ch/event/1109513/contributions/4969967/>. Used as conceptual motivation, not evidence of established extra dimensions.
- [2] G. Dusson, I. M. Sigal and B. Stamm, *The Feshbach–Schur map and perturbation theory* (2021), <https://arxiv.org/abs/2105.02058>.
- [3] B. Henning, X. Lu and H. Murayama, *One-loop Matching and Running with Covariant Derivative Expansion* (2016), <https://arxiv.org/abs/1604.01019>. The source-response algebra above is tree level, not their completed one-loop matching calculation.